

Part 1: Descriptives

Tasks:

1. Load the dataset into your environment.
2. Compute the following:
 - a. Mean
 - b. Median
 - c. Mode
 - d. Range
 - e. Variance
 - f. Standard Deviation
 - g. Interquartile Range (IQR)
 - h. Skewness
 - i. Kurtosis

Descriptive Statistics*Descriptive Statistics*

CPU Temperature	
Mode	47.000 ^a
Median	47.900
Mean	47.673
Std. Deviation	2.780
IQR	4.800
Variance	7.728
Skewness	-0.255
Std. Error of Skewness	0.427
Kurtosis	-1.029
Std. Error of Kurtosis	0.833
Range	9.900

^a The mode is computed assuming that variables are discreet.

Questions:

- **What does the mean tell you about the data?**

Based on the data, the CPU Temperature's mean is 47.673, which summarizes the overall temperature of the dataset.

- **Is the median close to the mean? What does this suggest?**

Yes, the median is quite close to the mean with a 0.227 difference, which indicates that the data is symmetrical and not heavily skewed.

- **How spread out are the data?**

Given that the value of Standard deviation is 2.780, Interquartile Range (IQR) is 4.800, Range is 9.900, and Variance is 7.728, this only indicates that there are some variations on the data, but it's not that high.

- **Are there any indicators of skewness or outliers?**

For the skewness, it has a value of -0.255, which indicates that it's closer to 0, which also means that the data is symmetrical and there might be some data on the higher temperature end. For the outliers, the Kurtosis has a value of -1.029, which indicates

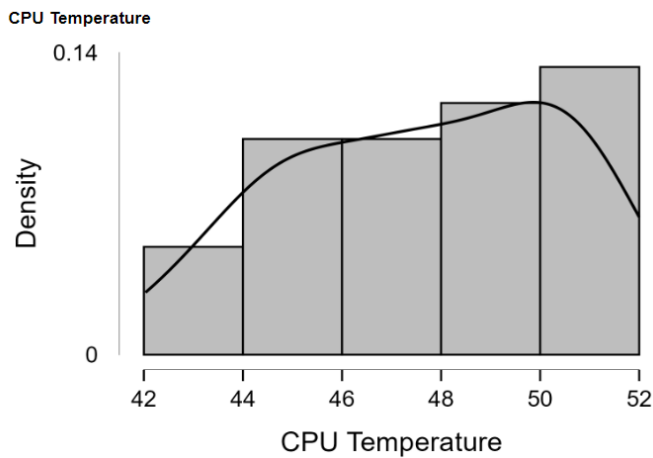
that the data has fewer outliers because having a negative kurtosis means that it has lighter tails and is flatter than a normal distribution.

Part 2: Graphical Representation

Tasks:

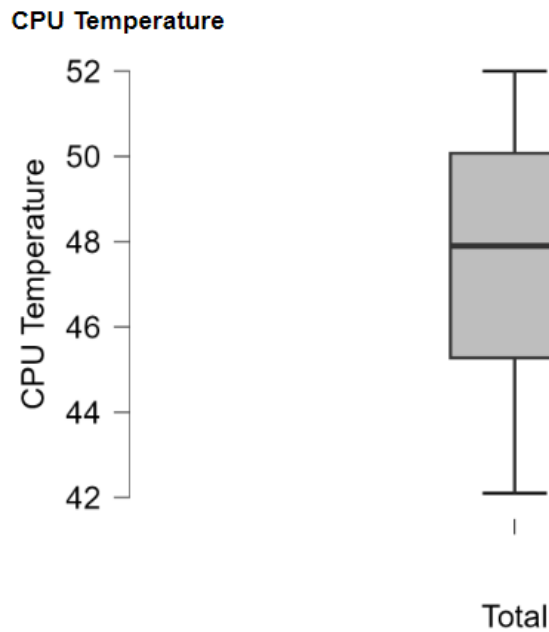
1. Create the following visualizations:

- Histogram (with KDE or smoothed curve)



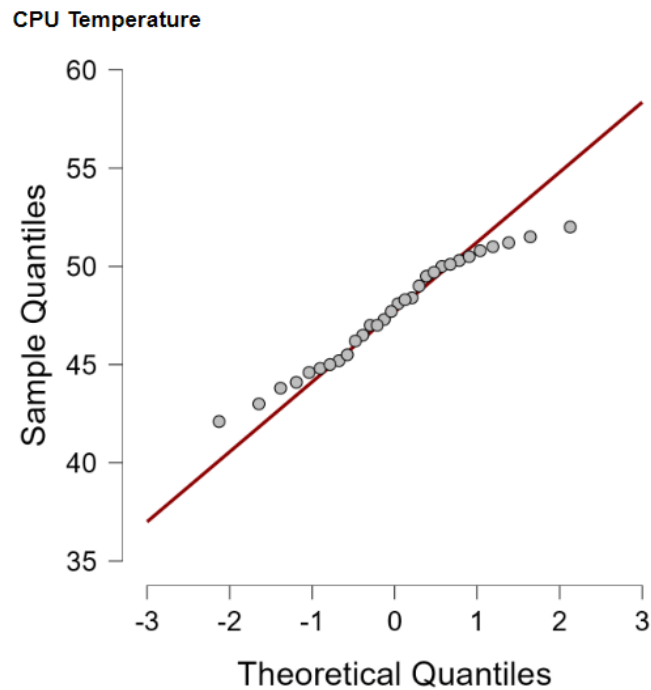
- Boxplot

Boxplots



- Q-Q Plot (for normality)

Q-Q Plot



2. Identify any outliers from the boxplot.

Based on the boxplot generated, there is a potential outlier in the lower whisker because it seems to extend more than the upper whisker.

Questions:

- **Based on the boxplot, how symmetric is the distribution?**

The boxplot is somewhat symmetric because the middle line inside the box is near the center of the box, but the box itself is placed more on the upper part of the line causing the lower whiskers to be longer than the upper whiskers.

- **Does the histogram support the numerical findings (mean/median/mode)?**

Based on the results, the histogram does support the numerical findings of the mean, median, and mode. For the mode, the tallest bar ranges from 50-52 while the value of the mode from the dataset is 47.000, which means that the histogram is not directly

supporting the numerical findings for mode. For the median, the histogram supports the numerical findings for it because if the middle bar, which ranges from 46-48, is divided into 2, this will result in roughly 47.9. Lastly, for the mean, since the result of the histogram slightly looks like a bell curve, then it would mean that the mean will be possibly located at the center of the bars, which supports the numerical findings for mean.

- **What does the Q-Q plot suggest about the normality of the data?**

Based on the Q-Q plot generated from the dataset, the result is not perfectly normal, but not also non-normal because majority of the points follows the red line but there are also some points where they are on the lower tail, and some are on the upper tail.

Part 3: Interpretation and Summary

Tasks:

1. Analyze the distribution:

- **Is the temperature curve symmetrical or skewed?**

The temperature curve is not balanced because it goes up quickly and comes down slowly based on the previous dataset results. It leans more on the other side, which indicates that it's skewed.

- **Are there any indications of thermal throttling or overheating?**

There is a potential indication that thermal throttling occurs. Specifically, the system might have started to reduce its performance when it reached a thermal limit to prevent further overheating. However, upon reaching the peak temperature, there might also be a possibility that it will overheat if it's sustained.

- **How does the temperature rise and fall across the time frame?**

At the beginning, the temperature goes up fast and then it hits a peak and goes down more slowly over time after reaching its peak.

2. Summary Report:

- **Write a short paragraph explaining your observations.**

- **Include interpretation of visual graphs and numerical stats.**

The CPU temperature data shows a dynamic pattern over time, which includes the rapid increases of a certain temperature peak, followed by a more gradual

cooling period. As indicated in the histogram, the distribution of temperatures shows a slight positive skew, while the Q-Q plot suggests a deviation from almost perfect normality, with lighter lower tails and heavier upper tails. All in all, the CPU reached a peak or high temperature, following a cooling period. The results are somewhat or almost normal but there are still outliers present that can be seen in the visual graphs generated from the datasets.